

Assessment

Program : - CompTia N+

Assessment Code :- M2-A1

Total Time :- 3 Hours

SECTION – A CONCEPT APPLICATION

SCENARIO :- Nexus Innovations is purchasing new computers for employees who use programming software, web browsers, and video conferencing applications at the same time.

QUESTION :- What factors should be considered when selecting the CPU, RAM, and storage to ensure good performance and hardware compatibility?

- To ensure good performance and hardware compatibility for employees running programming software, web browsers, and video conferencing applications simultaneously at Nexus Innovations, consider the following factors:
- CPU (Processor):- Core Count and Threads: Programming environments (like IDEs and compilers) and video conferencing require multi-core processors. Look for CPUs with at least 6 to 8 cores (such as Intel Core i5/i7 or AMD Ryzen 5/7) to handle multitasking smoothly without lag during video calls.
 - Clock Speed: Higher clock speeds ensure that compiling code and executing web browsers with multiple tabs open remain snappy.

- RAM (Memory):- Capacity: Running heavy IDEs, multiple resource-heavy browser tabs, and Zoom/Teams calls simultaneously demands a minimum of 16GB of RAM, with 32GB recommended to future-proof the machines and prevent performance bottlenecks.
 - Speed and Type: Ensure the motherboard is compatible with modern high-speed RAM standards (e.g., DDR5 or DDR4) to handle data transfer efficiently.
- Storage:- Type: Use NVMe SSDs instead of traditional HDDs. SSDs drastically reduce application load times, project build times, and boot up sequences.
 - Capacity: At least 512GB (preferably 1TB) is essential to accommodate large software packages, source code repositories, virtual machines, and operating system updates.
- Hardware Compatibility:- Motherboard and Socket: Ensure the chosen CPU matches the motherboard's socket type and generation.
 - Power Supply (PSU) and Thermal Design: Verify that the power supply can adequately support the processor and other components, and that adequate cooling solutions are in place to handle sustained heavy loads during compilation.

SCENARIO:- After installing antivirus software on all startup office computers, employees notice that their systems have become slower.

Question:- Explain how the antivirus software affects CPU, RAM, and storage performance, leading to system slowdowns.

- Based on the scenario where office computers became slower after installing antivirus software, here is how it affects CPU, RAM, and storage performance:
- 1. CPU Performance Impact :-
- Background Scanning & Real-Time Monitoring: Antivirus software continuously runs background processes to intercept and inspect files, downloads, and running applications in real time.
- Resource Consumption: Every time an employee opens a file, runs a program, or browses the web, the CPU has to dedicate significant processing cycles to analyze code signatures and perform heuristic analysis. This extra computational overhead

leaves fewer CPU resources available for regular office applications, causing noticeable lag.

- 2. RAM (Memory) Impact :-
- Resident Services & Signature Databases: Antivirus programs keep active monitoring agents loaded into the system's RAM at all times.
- Memory Pressure: To scan files quickly, the software loads extensive virus definition databases and signature libraries directly into memory. This reduces the amount of free RAM available for other office programs (like web browsers, Excel, or email clients), leading to frequent swapping and system slowdowns.
- 3. Storage (HDD/SSD) Performance Impact :-
- Constant File Reads & Disk I/O: During full system scans or scheduled checks, the antivirus utility reads thousands of files from the storage drive sequentially or randomly.
- Storage Bottleneck: If the office computers use older HDDs (Hard Disk Drives), the constant mechanical read/write head movement creates a massive I/O bottleneck, spiking disk usage to 100%. Even on SSDs, intense parallel file scanning causes heavy write/read queues, delaying how fast regular applications can fetch their required data.

SCENARIO :- A USB keyboard and printer work properly on one computer but do not work correctly on another computer in the startup office.

Question :- Describe the steps you would follow to troubleshoot the problem if it is caused by driver incompatibility or operating system issues.

- To troubleshoot driver incompatibility or operating system issues causing the USB keyboard and printer to fail on the second computer, follow these steps in order:
- 1. Check Device Manager for Driver Errors Diagnostic Step :-
- Press the Windows key, type Device Manager, and open it. Look for your keyboard and printer under their respective categories or under "Other Devices" / "Unknown Devices" (which will show a yellow triangle or exclamation mark).
- Verification: The target devices are listed without any error codes (such as Code 10 or Code 43) or warning symbols.
- 2. Update or Reinstall the Driver Fix :-
- Right-click the problematic device in Device Manager, select Update driver, and choose "Search automatically for drivers." If that fails, download the latest official drivers directly from the manufacturer's website and install them. Alternatively, right-

click the device, select Uninstall device, and restart the computer so Windows can reinstall clean default drivers.

- Verification: The device status in Device Manager updates to "This device is working properly" and the manufacturer's control software recognizes it.
- 3.Run Built-in Windows Troubleshooters OS Diagnostics :-
- Open Settings > System > Troubleshoot > Other troubleshooters. Run the Keyboard troubleshooter and the Printer troubleshooter to automatically detect and repair corrupt registry entries or underlying service issues.
- Verification: The troubleshooter completes the scan and displays a message stating that the issues have been fixed or the Print Spooler/USB stack has been reset successfully.
- 4.Install Pending Operating System Updates OS Patching :-
- Go to Settings > Windows Update and click Check for updates. Outdated Windows builds can lack necessary INF files and USB controller definitions needed to properly handshake with newer or specific peripherals.
- Verification: Windows reports that your system is fully up to date and a mandatory system reboot finishes cleanly.

SCENARIO :- Nexus Innovations experiences frequent power cuts and voltage fluctuations.

Question:- Explain how a UPS and surge protector help protect computer hardware and prevent data loss during these situations.

- In the case of **Nexus Innovations**, frequent power cuts and voltage fluctuations can damage computer hardware and cause sudden data loss. A **UPS (Uninterruptible Power Supply)** and a **surge protector** can help prevent these problems.
- A **UPS** provides temporary battery power when the main electricity supply fails. During a power cut, it immediately switches to battery power, giving employees enough time to **save their work and shut down the computers safely**. This prevents data loss, file corruption, and damage caused by sudden shutdowns. A UPS can also help maintain a stable power supply during minor voltage fluctuations.
- A **surge protector** protects computer components from sudden increases in voltage, known as **power surges**. It diverts excess voltage away from sensitive components such as the motherboard, power supply, hard disk, and other devices. This reduces the risk of hardware damage during voltage fluctuations or electrical disturbances.
- Therefore, Nexus Innovations should use **both a UPS and a surge protector**. The UPS mainly protects against **power cuts and unstable power**, while the surge protector protects against **sudden voltage spikes**. Together, they improve hardware safety, reduce downtime, and help prevent data loss.

SCENARIO :- New employees at Nexus Innovations report that they can reach internal systems (e.g. the application server) by IP address but not by name, and their devices sometimes fail to obtain a valid IP address when they connect to the network.

Question:- Explain how DNS and DHCP work together in this network, and identify which service is most likely misconfigured based on the symptoms described.

- In Nexus Innovations, DNS and DHCP work together to help employees connect to network resources easily.
- DHCP (Dynamic Host Configuration Protocol) automatically provides devices with important network settings such as an IP address, subnet mask, default gateway, and DNS server address. If DHCP is not working correctly, new employees' devices may fail to obtain a valid IP address and may not be able to communicate properly on the network.
- DNS (Domain Name System) converts human-readable names, such as `appserver.nexus.local`, into IP addresses. This allows employees to access the application server using its name instead of remembering its IP address.
- In this scenario, employees can reach the application server by IP address but not by name. This strongly indicates a DNS problem, because network connectivity by IP is working but name resolution is failing. However, because devices also sometimes fail to obtain a valid IP address, DHCP may also be misconfigured or unreliable.
- Therefore, DNS is the service most directly responsible for the name-resolution problem, while DHCP is likely responsible for the IP-address assignment issue. Both services should be checked, especially the DNS server settings being provided by DHCP.
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SCENARIO:- After the IT team installs a new firewall at Nexus Innovations, employees can access the internet but are unable to reach the critical internal application server.

Question:- Describe the troubleshooting steps you would take to determine whether the firewall rules are blocking legitimate traffic, and how you would resolve the issue.

- In **Nexus Innovations**, after installing the new firewall, employees can access the internet but cannot reach the **critical internal application server**. This suggests that the firewall may be blocking the traffic required to access the internal server.
- First, I would **check the firewall rules and policies** to see whether traffic from the employees' network to the application server's IP address and required port is being blocked. I would also verify the **source and destination IP addresses, ports, and protocols** in the rules.
- Next, I would check the firewall's **logs and blocked-traffic records** to identify whether connection attempts to the application server are being denied. I would test connectivity from an employee computer using tools such as **ping, tracert/traceroute, or Test-NetConnection** to determine where the connection is failing.
- If the firewall is confirmed to be blocking legitimate traffic, I would create or modify a **specific allow rule** permitting the required traffic from the trusted internal network to the application server. The rule should allow only the necessary **IP addresses, ports, and protocols**, rather than disabling the firewall completely.
- Finally, I would test the application from multiple employee computers and monitor the firewall logs to confirm that the issue is resolved. This approach keeps the firewall secure while allowing legitimate access to the critical application server.

SECTION B — PRACTICAL APPLICATION

Task 1: Workstation Assembly Physically assemble a workstation for the startup office and document the component installation sequence.

Install CPU, RAM, storage, and other core components into the chassis in the correct sequence. Connect all required internal cables (power, data, front panel). Document each installation step in order. System must pass POST (Power-On Self-Test) after assembly.

To physically assemble a workstation for the startup office by installing all major components correctly, connecting internal cables, and ensuring that the system successfully passes POST (Power-On Self-Test).

- Prepare the Work Area
Switch off the power supply and work on a clean, dry surface. Use an anti-static wrist strap or discharge static electricity before handling components.
- Install the CPU
Open the CPU socket on the motherboard and carefully place the processor according to the alignment mark. Lock the CPU socket securely.
- Install CPU Cooler
Apply thermal paste if required and install the CPU cooler over the processor. Connect the cooler's power cable to the CPU_FAN header on the motherboard.
- Install RAM
Open the RAM slot clips and insert the RAM module(s) into the recommended slots. Press firmly until the clips lock into place.
- Install Storage Drive
Install the SSD/HDD in its appropriate mounting position. For an M.2 SSD, insert it into the M.2 slot and secure it with the screw. For a SATA drive, connect the SATA data and power cables.
- Install the Motherboard in the Chassis
Fit the I/O shield if required, position the motherboard on the standoffs, and secure it using the appropriate screws.
- Install the Power Supply (PSU)
Mount the PSU inside the chassis and connect the required power cables, including the 24-pin motherboard power and CPU power connector.
- Connect Front-Panel Cables
Connect the Power SW, Reset SW, Power LED, and HDD LED cables to the correct motherboard front-panel headers.
- Connect Other Required Cables
Connect storage data cables, case fans, USB, and front-panel audio cables to their appropriate motherboard headers.
- Final Inspection
Check that all components are properly seated and that no loose screws or cables are interfering with fans or other components.
- Power On and Perform POST
Connect the monitor, keyboard, and power cable. Turn on the workstation. Check that the system starts normally and displays the POST/BIOS screen without error messages or warning beeps.

Task 2: RAM Mismatch Troubleshooting ASSESSMENT
FILE CompTia N+ M2-A1 Troubleshoot and resolve a RAM mismatch error within the BIOS/UEFI settings on a startup office workstation. Identify the RAM mismatch error via BIOS/UEFI POST messages.

Determine the root cause (mismatched speed, timing, or capacity). Resolve the mismatch by reconfiguring or reseating

the RAM correctly. Confirm the system boots normally after the fix

To identify and resolve a RAM mismatch error in the BIOS/UEFI of a startup office workstation and confirm that the system boots normally after the fix.

- Power On the Workstation
Start the workstation and observe the POST screen for any RAM-related error messages, warnings, or beep codes.
- Enter BIOS/UEFI
Press the appropriate key, such as Delete, F2, or F10, during startup to enter the BIOS/UEFI settings.
- Check RAM Information
Open the memory or system information section and check the installed RAM's:
 - o Total capacity
 - o Speed/frequency
 - o Timing
 - o Number of installed modules
- Identify the Root Cause
Compare the installed RAM modules. A mismatch may occur because the modules have different capacities, speeds, or timings. For example, one RAM module may operate at 3200 MHz while another is rated for 2666 MHz.
- Reseat the RAM
Shut down the workstation and disconnect the power. Remove the RAM modules and carefully reinstall them in the motherboard's recommended slots, ensuring that they are fully locked into place.
- Reconfigure BIOS/UEFI
If the RAM modules are compatible but have different speeds or timings, set the memory configuration to a common supported speed/timing in BIOS/UEFI. Avoid using settings higher than the motherboard or RAM supports.
- Test the System
Save the BIOS/UEFI settings and restart the workstation. Check whether the RAM mismatch warning has disappeared and whether the system completes POST successfully.
- Confirm Normal Boot
Allow the operating system to start and verify that the full expected RAM capacity is detected and the workstation operates normally.

Task 3: LAN Topology Build Build a LAN topology in Packet Tracer for 5 systems and 1 printer using a Class C subnet. Use Cisco Packet Tracer to build the topology. Include 5 end-systems and 1 network printer.

Assign IP addresses from a Class C subnet to all devices. All assigned IPs must be reachable via Ping

- To build a LAN topology in Cisco Packet Tracer with 5 systems and 1 network printer using a Class C subnet, assign IP addresses to all devices, and verify connectivity using Ping.
- Devices Required
- 1 × Cisco Switch (2960)
- 5 × PCs
- 1 × Network Printer
- 6 × Copper Straight-Through cables
- Network Details
- Network: 192.168.1.0/24
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1
- Device - IP Address - Subnet Mask
- PC0 - 192.168.1.2 - 255.255.255.0
- PC1 - 192.168.1.3 - 255.255.255.0
- PC2 - 192.168.1.4 - 255.255.255.0
- PC3 - 192.168.1.5 - 255.255.255.0
- PC4 - 192.168.1.6 - 255.255.255.0
- Printer - 192.168.1.7 - 255.255.255.0
- Topology
- Cisco 2960 Switch
- / / | \ \
- / / | \ \
- PC0 PC1 PC2 PC3 PC4
-
- |
- Printer
- Configuration Steps
- Open Cisco Packet Tracer.
- Add 1 Cisco 2960 switch.
- Add 5 PCs and 1 Printer.
- Connect every device to the switch using Copper Straight-Through cables.
- Configure the IP addresses:
- Click PC → Desktop → IP Configuration.
- Enter the assigned IP address.

- Enter subnet mask 255.255.255.0.
- Configure the printer:
- Click Printer → Config → FastEthernet.
- Set IP address to 192.168.1.7.
- Set subnet mask to 255.255.255.0.
- Ping Verification
- From PC0, open Desktop → Command Prompt and enter:
- ping 192.168.1.3
- ping 192.168.1.4
- ping 192.168.1.5
- ping 192.168.1.6
- ping 192.168.1.7
- The result should show:
- Reply from 192.168.1.x
- This confirms that all 5 systems and the printer are reachable on the LAN.
- Result
- The LAN topology was successfully created in Cisco Packet Tracer using a Class C subnet 192.168.1.0/24. Five PCs and one network printer were assigned unique IP addresses, and connectivity was successfully verified using Ping.

Task 4: Clean OS Installation Perform a clean OS installation with manual disk partitioning and startup optimization on a startup office workstation. Perform a clean install of Windows/Linux from ISO.

Manually partition the disk (not guided/automatic partitioning). Optimize startup programs/services for faster boot. OS must boot without errors after installation

- To perform a clean installation of an operating system using an ISO file, manually partition the disk, and optimize startup programs/services to improve boot time.
- Requirements
- Startup office workstation
- Windows/Linux ISO file
- Bootable USB drive
- Minimum required disk space
- Keyboard and mouse
- Procedure
- 1. Create Bootable USB

- Download the required Windows/Linux ISO.
 - Create a bootable USB drive using a tool such as Rufus.
 - Connect the USB drive to the workstation.
 - 2. Boot from USB
 - Restart the computer.
 - Open the Boot Menu/BIOS using the appropriate key such as F12, F2, Del, or Esc.
 - Select the USB drive as the boot device.
 - Start the OS installation.
 - 3. Perform Clean Installation
 - Select Install Now.
 - Choose the required edition and accept the license terms.
 - Select Custom: Install Windows only for Windows.
 - Delete old OS partitions only after confirming that important data has been backed up.
 - 4. Manual Disk Partitioning
 - Example for a 256 GB disk:
- | | | |
|--------------|-------------------|----------------------|
| - Partition | - Suggested Size | - Purpose |
| - System/EFI | - 100–260 MB | - Boot files |
| - MSR | - 16 MB | - Microsoft Reserved |
| - C: | - 150 GB | - Operating System |
| - D: | - Remaining space | - Data |
- Select the appropriate partition for the OS and continue the installation.
 - 5. Complete OS Installation
 - Allow the installation to copy files and restart the computer.
 - Complete the initial setup.
 - Install required drivers and Windows/Linux updates.
 - Confirm that the desktop loads normally without errors.
 - Startup Optimization
 - For Windows:
 - Press Ctrl + Shift + Esc to open Task Manager.
 - Select Startup apps.
 - Disable unnecessary applications that automatically start with Windows.
 - Do not disable essential Windows, security, hardware, or driver services.
 - Restart the computer and check the boot time.
 - Verification
 - After installation and optimization:
 - System boots successfully.
 - No boot errors are displayed.
 - OS loads to the desktop normally.
 - Disk partitions are visible and correctly configured.
 - Required drivers work correctly.

- Unnecessary startup applications are disabled.
- System starts faster and is ready for office use.

SECTION C — MINI CAPSTONE PROJECT

Mini Project: Nexus Innovations — Diagnosing Random System Reboots Objective: At Nexus Innovations, three rendering workstations pass POST every morning but restart randomly during heavy afternoon workloads. The computers are kept inside poorly ventilated desk cabinets. Analyze the probable hardware and thermal causes and propose a remediation plan, combining hardware, power, and troubleshooting concepts from this module.

Your project must: Identify the three hardware components most likely to cause random reboots under load. Explain why the systems pass POST in the morning but fail in the afternoon. Describe how you would use a stress-test tool to confirm a thermal or power issue. Propose three specific steps to remediate the frequent system restarts. Provide a brief technical justification for why a PSU upgrade might (or might not) be necessary.

- To diagnose the cause of random system reboots in three rendering workstations and propose a practical remediation plan based on hardware, thermal, and power-related troubleshooting concepts.
- 1. Three Hardware Components Most Likely to Cause Random Reboots
- 1. CPU (Processor)
Rendering workloads heavily use the CPU and generate significant heat. If the CPU overheats, the system may automatically shut down or restart to protect the hardware.

- 2. GPU (Graphics Card)
Rendering places a heavy load on the GPU. A high GPU temperature or insufficient cooling can cause instability and unexpected restarts.
- 3. PSU (Power Supply Unit)
During heavy workloads, the CPU and GPU consume much more power. An aging, low-capacity, or faulty PSU may fail to provide stable power, causing sudden reboots.
- 2. Why the Systems Pass POST in the Morning but Fail in the Afternoon
- POST (Power-On Self-Test) mainly checks whether essential hardware is detected and functioning during startup. In the morning, the workstations are relatively cool and have not yet been subjected to sustained heavy workloads.
- In the afternoon, rendering workloads increase CPU and GPU utilization. Because the computers are kept inside poorly ventilated desk cabinets, hot air accumulates around the systems. As temperatures increase, CPU/GPU thermal protection may activate, or the PSU may become less stable under sustained load. This explains why the systems can pass POST but later restart during rendering.
- 3. Using a Stress-Test Tool
- I would first monitor the CPU and GPU temperatures, clock speeds, and power usage at idle. Then I would run a controlled CPU/GPU stress test such as OCCT or another reputable hardware stress-testing tool.
- Procedure:
- Record idle CPU/GPU temperatures.
- Start the stress test.
- Monitor temperatures and power consumption continuously.
- Run the test long enough to reproduce the problem.
- If temperatures rise excessively and the system restarts, a thermal problem is likely.
- If temperatures remain within safe limits but the system suddenly loses power/restarts under combined CPU/GPU load, a PSU or power-delivery problem becomes more likely.
- 4. Three Specific Remediation Steps
- Step 1 – Improve Airflow
- Remove the workstations from enclosed desk cabinets.
- Ensure adequate space around air intakes and exhausts.
- Clean dust from fans, heatsinks, and filters.
- Improve room ventilation.
- Step 2 – Service the Cooling System
- Check that CPU and GPU fans are operating correctly.
- Clean the heatsinks and fans.
- Replace degraded thermal paste if necessary.
- Verify CPU/GPU temperatures under load after servicing.
- Step 3 – Check and Correct Power Delivery
- Check PSU wattage, condition, and power connectors.
- Test the workstation under CPU + GPU load.
- Replace the PSU if it is under-rated, unstable, damaged, or failing under load.

- 5. Is a PSU Upgrade Necessary?
- A PSU upgrade may or may not be necessary. It should not be replaced simply because the system is restarting.
- If testing shows that the PSU has sufficient wattage, is in good condition, and provides stable power under maximum CPU/GPU load, the main problem is more likely to be thermal, especially because the workstations are located in poorly ventilated cabinets.
- However, if the PSU is under-rated, old, faulty, or unable to provide stable power during peak rendering loads, upgrading to a quality PSU with sufficient wattage and appropriate power connectors would be necessary.
- Conclusion
- The most probable causes of the random reboots are CPU overheating, GPU overheating, and PSU/power instability. The enclosed cabinets worsen heat buildup during afternoon rendering workloads. The recommended solution is to improve airflow, service the cooling system, monitor temperatures during stress testing, and replace the PSU only if power testing confirms that it is inadequate or unstable.

SECTION D – AI AUGMENTED LEARNING

STEP 1 · BUILD WITH AI Use an AI tool of your choice (ChatGPT, Claude, GitHub Copilot, etc.) to generate an IP addressing and subnetting plan for Nexus Innovations' startup office network, covering 5 systems and 1 printer. The AI-generated plan must specify the subnet, subnet mask, usable host range, and broadcast address. The plan must allocate addresses for all 5 systems and 1 printer named in the scenario. Instruct the AI to also identify which addresses to reserve for future expansion. Document the exact prompt(s) used.

STEP 2 · TEST & DEBUG (WITHOUT AI) Then, working without AI, verify the plan for errors —

overlapping ranges, incorrect network/broadcast addresses, insufficient host capacity, or unreserved growth space — and correct them yourself.

- Step 1: Build with AI (AI-Generated Plan)
- Prompt Used:
- "Generate an IP addressing and subnetting plan for Nexus Innovations' startup office network consisting of 5 systems and 1 printer. The plan must specify the network subnet, subnet mask, usable host range, and broadcast address. Allocate specific static IP addresses for the 1 printer and 5 client systems, and explicitly reserve a block of addresses for future office expansion."
- AI-Generated Network Plan:
- Network Subnet: 192.168.10.0
- Subnet Mask: 255.255.255.240 (CIDR: /28, total 16 IP addresses)
- Usable Host Range: 192.168.10.1 to 192.168.10.14
- Broadcast Address: 192.168.10.15
- Initial IP Allocation Table:

- Device Name / Role	- IP Address	- Type / Status
- Default Gateway / Router	- 192.168.10.1	- Static (Infrastructure)
- Office Printer	- 192.168.10.2	- Static (Assigned)
- System 1	- 192.168.10.3	- Static (Assigned)
- System 2	- 192.168.10.4	- Static (Assigned)
- System 3	- 192.168.10.5	- Static (Assigned)
- System 4	- 192.168.10.6	- Static (Assigned)
- System 5	- 192.168.10.7	- Static (Assigned)
- Future Expansion	- 192.168.10.8 to 192.168.10.14	- Reserved (7 IPs available)

- Step 2: Test & Debug (Manual Verification & Correction)
- Working without AI, the generated plan is verified against potential errors:
- Overlap Check: The subnet 192.168.10.0/28 is self-contained and does not overlap with standard home/office router defaults (like 192.168.1.1 or 192.168.0.1), preventing routing conflicts.
- Network & Broadcast Address Accuracy:
- Network address (192.168.10.0) and broadcast address (192.168.10.15) are correctly excluded from host assignment.
- Host Capacity & Growth Space:
- Total usable hosts in a /28 mask = $2^4 - 2 = 14$ addresses.
- Current active devices = 6 (5 systems + 1 printer) + 1 gateway = 7 addresses utilized.
- Remaining growth space = 7 reserved addresses (.8 through .14), which provides adequate room for immediate startup expansion without requiring immediate subnet redesign.
-